Justin Wu

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Education

Carnegie Mellon University Pittsburgh, PA
Bachelor of Science in Artificial Intelligence *August 2026 – May 2029*
GPA: 4.0/4.0

Experience

Research Assistant June 2025 – August 2026
Nathanson Lab, Perelman School of Medicine, University of Pennsylvania *Philadelphia, PA*

- Conducted association testing on approximately 500K single-nucleotide polymorphisms (SNPs) across thousands of samples in a genome-wide association study (GWAS) meta-analysis of polygenic risk scores and cancer relapse in testicular germ cell tumors, producing Manhattan and QQ plots to assess genome-wide significance
- Engineered a Bash/Python pipeline to identify proxy SNPs in linkage disequilibrium ($r^2 \geq 0.8$) with 100+ lead GWAS hits across 500 kb windows, using LocusZoom for regional LD visualization to enable downstream fine-mapping
- Harmonized SNP naming and genome builds across the Penn Medicine BioBank (PMBB), Gencove, and Discovery cohorts, correcting an outdated marker list and tripling the size of the shared replication set

Projects

Apex June 2025 – June 2026
3D F1 Race Replay Engine | Three.js, Vite, FastF1

- Built an interactive 3D replay engine visualizing ∼140 Formula 1 races since 2020 from real FastF1 telemetry, rendering ∼250K data points per race at a sustained 60 FPS
- Applied Kalman filtering to smooth noisy telemetry, cutting car jitter by 50%, and physics-informed splines to interpolate car paths without overlap between cars
- Implemented multiple camera modes (free orbit, follow car, TV chase/action cams) and variable-speed playback (0.5x–64x) synchronized with telemetry

Publications & Presentations

Enhancing Genetic Ancestry Inference through TDA and Unsupervised Clustering July 2026
Wu, J., & Yang, T. — Poster, Intelligent Systems for Molecular Biology (ISMB) Conference

- Paired topological features with unsupervised clustering to improve ancestry inference in admixed populations, reaching 96% accuracy on the admixed American (AMR) superpopulation versus 73% for PCA with k-means
- Applied persistent homology to 1000 Genomes Project SNP data from 2,504 individuals across 26 populations, generating persistence barcodes to capture population structure beyond what linear methods such as PCA reveal
- Maintained overall ancestry classification accuracy of 99.08% using topological data analysis (TDA) with Ward linkage clustering, on par with PCA with k-means (99%)

Technical Skills

Languages: Java, Python, C/C++, JavaScript, HTML/CSS, R, Swift
Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, Eclipse, Bash, npm
Libraries: pandas, NumPy, Matplotlib, SciPy, scikit-learn, seaborn, statsmodels, Biopython

Honors & Awards

- 2x American Invitational Mathematics Examination (AIME) Qualifier (2023, 2025)
- USA Biolympiad, Recognition of Academic Excellence — top 50 of 10000+ competitors (2025)
- MathWorks Math Modeling (M3) Challenge, Honorable Mention — top 25 of 800+ papers (2025)
- Future Business Leaders of America (FBLA) PA State Champion, Computer Problem Solving (2025)
- Pennsylvania Seal of Biliteracy, Spanish (2026)